

Ryan Cho

Senior Software Engineer | Test Automation & Quality Engineering

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PROFESSIONAL SUMMARY

Software engineer with 8+ years testing complex web applications across every SDLC phase, from exploratory validation through automated regression at scale. Deep expertise in Selenium, TestNG, and Java automation, advanced SQL backend verification, and JMeter performance testing. Delivered document validation with Apache POI and PDFBox, WebTable parsing for dynamic UIs, and Zephyr Scale/JIRA workflows at financial institutions and SaaS platforms.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and maintained a **Selenium and TestNG automation framework** spanning 400+ regression scenarios across mission-critical web application workflows, cutting manual QA effort by 60% each release cycle.
- Led **end-to-end SDLC quality assurance** for SaaS platform features—requirements review, test planning, execution, and defect triage—partnering with product owners and engineering leads on every sprint.
- Designed **Java validation libraries** using Apache POI and Apache PDFBox to automate spreadsheet and PDF output verification, eliminating hours of manual reconciliation per reporting period.
- Built **JMeter performance test suites** simulating up to 5,000 concurrent user sessions, surfacing throughput bottlenecks and latency regressions before production deployments.
- Engineered **advanced SQL validation queries** against PostgreSQL and Oracle backends to confirm transactional integrity, report accuracy, and cross-system data consistency.
- Implemented **WebTable parsing utilities** within automation scripts to dynamically extract, sort, and assert paginated grid data across administrative web interfaces.
- Integrated **Zephyr Scale and JIRA** into the QA workflow, standardizing test case management, requirement traceability, and release sign-off documentation across distributed engineering teams.
- Mentored three engineers on **test automation standards**, flaky-test root-cause analysis, and code review practices, raising team ownership of pre-production quality gates.
- Collaborated with architects on **cross-team test strategy** spanning API and UI layers, influencing CI/CD pipeline quality gates that blocked defective builds from reaching staging.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Owned **functional and regression testing** for complex internal banking web applications across trade settlement, risk reporting, and client portal modules throughout the full SDLC.
- Developed **Selenium WebDriver automation in Java** with TestNG parallel execution, expanding automated regression coverage from 30% to 75% over 18 months.
- Authored **advanced SQL scripts** for database state validation, reconciliation reporting, and backend verification supporting nightly batch processing checkpoints.
- Executed **performance and load testing with JMeter** against critical APIs and web endpoints, documenting latency thresholds and capacity limits for production readiness reviews.
- Automated **spreadsheet and PDF output verification** with Apache POI and Apache PDFBox, validating regulatory report formats against business-defined schemas.
- Parsed **dynamic WebTables and data grids** within automated test flows to confirm sort order, pagination, and filter behavior on portfolio management screens.
- Managed defect lifecycle and **test execution tracking in JIRA**, coordinating with offshore QA analysts and development leads to resolve blockers before release deadlines.

- Contributed **test case design in Zephyr Scale**, linking requirements to executable scenarios and maintaining audit-ready traceability for compliance reviews.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Supported **manual and automated testing** of a React-based web application, executing test cases across Chrome and Firefox during active sprint cycles.
- Built initial **Selenium scripts in Java** with TestNG for login, checkout, and account management flows, establishing the team's first repeatable regression suite.
- Wrote **SQL queries** to validate user records and transaction logs in MySQL after UI test runs, catching data persistence defects before staging promotion.
- Assisted senior engineers with **SDLC test planning**—reviewing user stories, identifying edge cases, and documenting reproduction steps in JIRA.
- Participated in **cross-browser compatibility testing** for form submissions and dynamic content rendering across desktop and mobile viewports.
- Helped configure **CI-triggered test runs** on pull requests, reducing undetected regressions reaching the staging environment.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Conducted **manual testing of web application features** including navigation flows, form validation, and responsive layout checks before each client demo.
- Documented defects and enhancement requests in **JIRA**, providing clear reproduction steps and screenshots for developer handoff.
- Executed **smoke and regression test passes** across the SDLC for a nonprofit platform, verifying fixes before production releases.
- Learned foundational **Selenium automation concepts** by pairing with senior contributors on scripted login and search workflow tests.
- Validated **backend data displays** on administrative dashboards with basic SQL lookups, ensuring UI values matched database records.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: HTML5, CSS3, JavaScript, React, WebTable DOM parsing, cross-browser testing, responsive UI validation

Backend: Java, Selenium WebDriver, TestNG, REST API testing, Apache POI, Apache PDFBox, Maven

Databases: SQL (advanced queries, joins, stored procedures), PostgreSQL, MySQL, Oracle, data reconciliation

Cloud & DevOps: Jenkins, GitHub Actions, Docker, CI/CD test pipelines, AWS (EC2, S3)

Monitoring & Tools: JMeter, WAPT Pro, Selenium Grid, Zephyr Scale, JIRA, Postman, BrowserStack

Workflow & PM: SDLC testing, Agile/Scrum, test case management, defect triage, stakeholder reporting, Zephyr Scale